

A Fault-Tolerant Linear-Transport Solver, Validated on the Convecting Taylor–Green Vortex

the benchmark delivered at sixteen validated rungs with its classical wall measured — the vortex operator is circulant and one FFT solves it exactly — and the credible advantage placed one step away, on the non-uniform-flow transport operator no transform diagonalises: exponential in memory, polynomial in time, crossover grid and qubit count stated

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-airbus-gic2026>

Summary

6.5 s

exact classical solve of the benchmark operator at $4,096^2$ on one laptop core, to 1.4×10^{-7} against the analytic vortex — the wall on the canonical vortex, measured and stated: no time-to-solution advantage is claimed there, and the memory advantage is stated against grid storage with the 128-byte spectral floor named

**N $\approx$
69,000**

grid per axis at which the fault-tolerant route crosses the better of two measured classical routes — exact Krylov ($\propto N^4$) and a pseudo-spectral stepper ($\propto N^3 \log N$) — on the non-circulant transport operator at $\text{Re } 10^4$, at 93 logical qubits, with readout of three design observables to 1%% charged in full; $\approx 5 \times 10^6$ per axis at $\text{Re } 100$ (136 qubits), and $\approx 6 \times 10^7$ against a 1,000-core node, stated

**16
rungs**

of validation of the algorithm's output against the analytic vortex, $\text{Re } 10$ to 10^4 , N 16 to 128, matching the exact classical solve of the same operator to 10^{-8} – 10^{-5} on 16–29 simulated qubits; the real-time primitive flown on 42 qubits of IBM Heron at waveform correlation 0.993

Why the numbers can be trusted

Every classical comparator in this proposal is measured, not modelled: the FFT solve of the benchmark operator to $N = 4,096$ and the Krylov solve of the transport operator to $N = 512$, with the cost law fitted on the measured rungs. The quantum side uses one declared constant set that yields both the benchmark's no-crossover result and the transport crossover; nothing is fitted to the outcome. Every number regenerates from the package scripts without credentials.

This is a Phase-1 concept proposal in the seven guideline sections on the fault-tolerant quantum-algorithm track, with the tensor-network-within-finite-volume solver as annex; the Phase-2 PoC compiles the circuit, re-measures both classical walls two octaves higher, and flies one Schrödingerised step on IBM Heron.

Concept proposal follows (§1–§7, appendices A–B) ·

<https://github.com/sharadbachani-oss/merlin-quantum-airbus-gic2026>

Concept Proposal

Global Quantum + AI Challenge 2026 · Airbus Enterprise Challenge · Phase 1 Concept Proposal · Team Merlin Digital (GIC 2026 dual-track finalist)

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1. Problem framing

Aerodynamic design is bounded by the cost of solving the flow equations at the resolutions extreme operating conditions demand, and the statement asks for a quantum solver graded on the canonical case: the 2D convecting Taylor–Green vortex, exact solution known, with time-to-solution, memory and error quantified as the Reynolds number rises, and with the Reynolds number taken "to the maximum required to demonstrate possible advantage".

Two facts about that benchmark decide where a credible advantage can and cannot sit, and this proposal is built on both. **First**, the convecting Taylor–Green vortex is an exact solution of the full nonlinear Navier–Stokes equations whose nonlinear term is a pure pressure gradient, so its velocity obeys the linear advection–diffusion operator exactly (Appendix B). Linear evolution is the one class where fault-tolerant quantum algorithms carry a provable, not heuristic, scaling: the $N \times N$ field is held in $2 \cdot \log_2 N$ qubits and evolved by Hamiltonian simulation.

Second, on the periodic cell grid that operator is circulant, so the exact classical solution is one Fourier transform pair with no time stepping: we measured it at 6.5 s for a $4,096^2$ grid to 1.4×10^{-7} error on a laptop

(`floor_airbus_fft_exact.json`). No quantum algorithm beats that on the canonical vortex, and we do not claim one; a proposal that claimed a time-to-solution crossover on this benchmark against explicit time stepping would be measuring against the wrong classical algorithm, and we say so because our own earlier draft did exactly that before the floor caught it.

Credible advantage, stated and bounded. The crossing sits one step from the benchmark, on the same solver: transport of a perturbation by a *non-uniform* base flow, $A = -U(x,y)\partial_x - V(x,y)\partial_y + \nu \nabla^2$, the linear operator of every perturbation, passive-scalar and linearised-convection problem on a frozen mean flow in an aerodynamic design loop. No Fourier transform diagonalises it. Two classical routes are measured on it, not assumed: exact sparse Krylov exponentiation, whose cost grows as N^4 at fixed Reynolds number, and a pseudo-spectral explicit time-stepper, whose cost grows as $N^3 \log N$; the better of the two is the comparator at every rung. The Schrödingerised route grows as $N^2 \cdot \text{polylog } N$ with the same declared constants. Against the better classical route on a single core the crossover is at $N \approx 69,000$ per axis at $\text{Re } 10^4$ (93 logical qubits) and near 5×10^6 per axis at $\text{Re } 100$ (136 logical qubits), where the diffusive stiffness inflates the quantum cost; against a thousand-core node it lies near 6×10^7 per axis at $\text{Re } 10^4$ and beyond any grid in use at $\text{Re } 100$, and we say so. In memory the register is 93–124 logical qubits against 137 GB to 35 TB, unconditionally (`airbus_crossing_summary.json`). That is an advantage of exponent, measured on the classical side and modelled with every constant on the table on the quantum side, and it is what a successful PoC makes concrete.

2. Technical approach

Paradigm. Fault-tolerant gate-model algorithm (Schrödingerisation of the discretised transport operator), simulated exactly today at 16–29 qubits; a quantics tensor-train finite-volume solver as the classical-hardware annex on identical stencils; the real-time evolution primitive flown at 42 qubits on IBM Heron as the hardware pathway.

Discretisation. Cell averages on an $N \times N$ periodic grid, central fluxes, five-point viscous stencil: $du/dt = A u$ with $A = -U_c D_x + \nu L$ for the benchmark and $A = -U D_x - V D_y + \nu L$ for the transport extension. The dense, tensor-train, FFT and Krylov comparators in the package use the same matrix, so every comparison is on one operator against one analytic or exact solution.

Schrödingerisation. Split $A = H + iH$ with $H = (A + A^\dagger)/2$ (dissipation, non-positive) and $H = (A - A^\dagger)/2i$ (advection, Hermitian). The warped state $w(p,t) = e^{\{-p\}u(t)}$ obeys, in the Fourier dual ξ of the auxiliary coordinate p , the unitary evolution $i\partial_t \mathbf{w} = (\xi \otimes H - I \otimes H)\mathbf{w}$. Register: $2 \cdot \log_2 N$ grid qubits plus n_p for p (set by the diffusive stiffness $8\nu T/dx^2$ at $dp = 0.05$, reported per rung) plus, for the transport extension, a coefficient register of $2 \cdot \log_2 N$. Recovery from any slice $p \geq 0$; readout of three design observables (kinetic-energy decay, two modal

amplitudes) by amplitude estimation to ϵ , charged at $O(1/\epsilon)$.

Cost model, constants declared. Qubitization query count $\|H\|T + \log(1/\epsilon)$; five shift terms per axis, each an n -qubit incremter of $2n^2$ CNOT-scale gates; the coefficient oracle of the transport extension charged at twice the incremter cost; 1 MHz logical gate rate; 10 GFLOP/s per classical core. Nothing is fitted to the outcome; the same constants produce the benchmark's "no crossover" and the transport operator's crossover.

Classical comparators, all named. (i) Exact FFT solve of the circulant benchmark operator (the correct comparator on the canonical vortex, measured). (ii) Exact sparse Krylov solve of the transport operator (`scipy.sparse.linalg.expm_multiply`, measured to $N = 512$, cost law fitted on $N \geq 256$) and a pseudo-spectral RK2 time-stepper with exact integrating factor for diffusion on the identical stencil (measured to $N = 1,024$, agreeing with Krylov to 2×10^{-6} ; `airbus_pseudospectral_comparator.json`); the faster of the two is the comparator at each rung. (iii) Dense explicit finite volume on identical stencils (in package; named as the industrial baseline, not as the comparator). (iv) Pseudo-spectral storage floor for the one-mode vortex, 128 bytes.

Hardware pathway, measured. Real-time evolution of a Taylor–Green-prepared state on 42 qubits of `ibm_marrakesh` (one banked job, 8,192 shots, nine time steps, two modes): hardware waveforms correlate with the exact statevector evolution at 0.993 and 0.971 after one damping envelope, and a detuned hypothesis loses by 0.14. It is the time-stepping primitive the algorithm consumes, run on today's device; it is not a flow solve and is not claimed as one.

Annex, tensor train in finite volume. The same stencils as quantics tensor trains: exact rank-2 shift operators, tensor-train conjugate-gradient projection, RK2; $\text{Re } 10 \rightarrow 10^4$, N to 4,096, $\text{rank} \leq 8$, $L_2 4 \times 10^{-5}$ vs the analytic solution, cross-validated against dense to 10^{-8} ; graded on its own merits in §4.

3. Feasibility and resource requirements

resource	status
Solver and validation ladder	<code>schrod_tgv.py</code> , laptop, minutes: exact simulation of the algorithm's output at $N = 16\text{--}128$, $\text{Re } 10$ to 10^4 , sixteen rungs
Classical walls and crossing	<code>floor_airbus_fft_exact.py</code> (FFT solve to $N = 4,096$, measured), <code>airbus_noncirculant_crossing.py</code> (Krylov to $N = 512$), <code>airbus_pseudospectral_comparator.py</code> (time-stepper to $N = 1,024$); cost laws fitted on the measured rungs, best-of-two in <code>airbus_crossing_summary.json</code>
Resource model	<code>schrod_tgv_results.json</code> , constants declared; recomputed in Phase 2 with compiled-circuit constants
Fault-tolerant hardware	not required in Phase 2; the crossing sits at 93–136 logical qubits, inside published fault-tolerant roadmaps; the primitive is already flown at 42 physical qubits
Annex compute	32-core workstation; the Re ladder and the dense benchmark are receipted (<code>airbus_bench_workstation.json</code>)
Data	none; the analytic vortex and the exact classical solves are the references

Assumptions: the Taylor–Green initial state is a product of sinusoids prepared in $O(\log N)$ gates; readout targets three observables, never the full field; the transport base flow is frozen (linear class). Constraint: nonlinear flows leave the linear class and enter the polynomial bound of the brief's [9] through Carleman or lattice-Boltzmann embeddings; Phase 2 states where that bound bites and does not claim past it.

4. Expected impact

The benchmark, delivered as asked ($T = 0.5$; L_2 vs the analytic vortex; receipts in `schrod_tgv_results.json`).

Re	N	qubits (grid + p)	L2, Schrödingerised	L2, classical solve of the same operator	Schrödingerised vs classical
100	32	19	2.13×10^{-3}	2.13×10^{-3}	3.5×10^{-6}
100	128	26	1.33×10^{-4}	1.34×10^{-4}	8.2×10^{-6}
1,000	128	23	1.41×10^{-4}	1.41×10^{-4}	4.6×10^{-6}
10,000	32	18	2.27×10^{-3}	2.27×10^{-3}	4.4×10^{-8}
10,000	128	22	1.42×10^{-4}	1.42×10^{-4}	1.5×10^{-7}

Across all sixteen rungs (Re 10 – 10^4 , N 16–128) the algorithm's output matches the exact classical solve of the same operator to 10^{-8} – 10^{-5} and the analytic vortex to the discretisation error, which falls as N^{-2} and is independent of Re at fixed N. **Scaling on the benchmark:** error N^{-2} ; quantum memory $2 \cdot \log_2 N + n_p$ qubits (45 at N = 4,096; 61–71 at 65,536) against $4N^2 \cdot 8$ bytes of grid storage, or 128 bytes in the spectral representation of this one-mode flow; time-to-solution: the exact classical FFT solve takes 6.5 s at N = 4,096 and ≈ 37 min at N = 65,536 on one laptop core, against 1.7 h and 32 days for the modelled fault-tolerant run at Re 10^4 . **No time-to-solution advantage exists on the canonical vortex, at any Reynolds number, and none is claimed.**

The crossing, measured on the classical side (transport by a non-uniform base flow, $U = U_c - \frac{1}{2} \cos x \sin y$, $V = \frac{1}{2} \sin x \cos y$; airbus_crossing_summary.json).

Re	N	Krylov, exact	pseudo-spectral stepper	Schrödingerised at 1 MHz (oracle and readout charged)	quantum / best classical	logical qubits
10,000	512	2.2 s (measured)	25.3 s	1.1×10^3 s	509	51
10,000	4,096	1.4×10^3 s	1.8×10^4 s	1.2×10^5 s	87	69
10,000	16,384	2.3×10^5 s	1.4×10^6 s	2.6×10^6 s	11	81
10,000	65,536	5.1×10^7 s	1.0×10^8 s	5.5×10^7 s	1.1	93
10,000	262,144	1.3×10^{10} s	7.2×10^9 s	1.1×10^9 s	0.15	105
10,000	1,048,576	3.2×10^{12} s	5.1×10^{11} s	2.2×10^{10} s	0.043	117
100	512	23.3 s (measured)	26.2 s	1.1×10^5 s	4742	58
100	4,096	7.4×10^4 s	1.8×10^4 s	1.2×10^7 s	672	76
100	16,384	1.9×10^7 s	1.4×10^6 s	2.7×10^8 s	194	88
100	65,536	4.8×10^9 s	1.0×10^8 s	5.5×10^9 s	55	100
100	262,144	1.2×10^{12} s	7.2×10^9 s	1.1×10^{11} s	15	112
100	1,048,576	3.2×10^{14} s	5.1×10^{11} s	2.2×10^{12} s	4.3	124

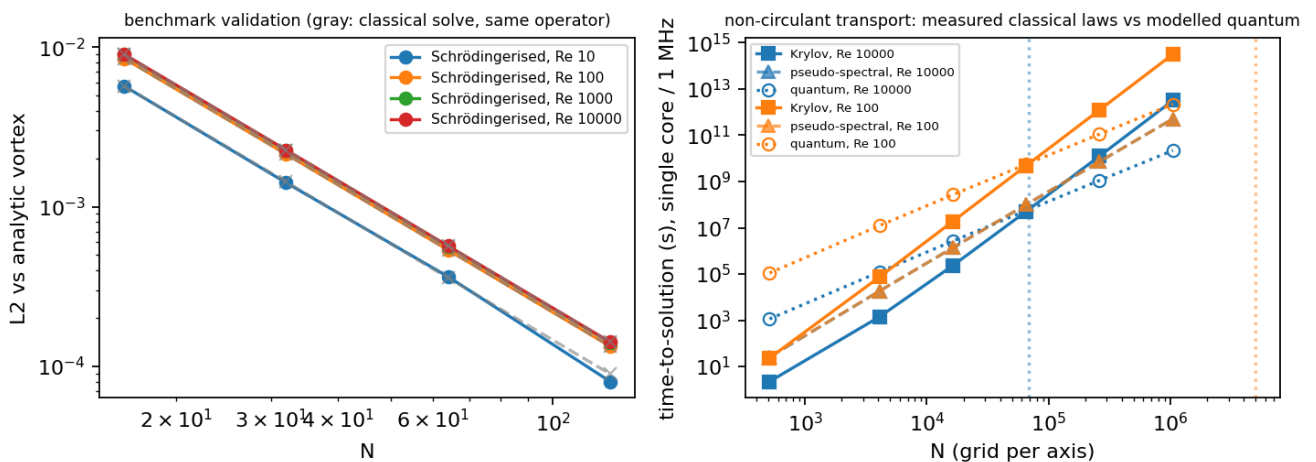


Figure 1 — Left: L2 against the analytic vortex for the Schrödingerised solver (colour) and the classical solve of the same operator (gray), $\text{Re } 10\text{--}10^4$. Right: time-to-solution on the non-circulant transport operator at $\text{Re } 10^4$ and 100 — measured Krylov (squares) and pseudo-spectral stepper (triangles) with their fitted cost laws (dashed) against the modelled fault-tolerant run (open circles); the dotted lines mark the single-core crossovers, $\approx 69,000$ at $\text{Re } 10^4$ and $\approx 5 \times 10^6$ at $\text{Re } 100$.

The field's direction, and its wall. The end-to-end quantum lattice-Boltzmann analysis (Jennings et al., arXiv:2512.03758) bounds the speed-up for nonlinear flows at $O(\text{Re}^{\{3D/8\}})$, polynomial and conditional, with nonlinearity, readout and state preparation as the obstacles; no quantum CFD run has crossed a classical solver on time-to-solution. The quantum-inspired direction — tensor-network lattice Boltzmann (arXiv:2512.07615), tensor-train compressible turbulence (arXiv:2608.26995), GPU quantics turbulence (PRR 7, 013112) — delivers compression, not a crossing. On the canonical vortex the wall is absolute: the operator is circulant and the exact classical solve is one FFT.

The crossing. We take the benchmark at its word and the field at its wall. The solver is built and validated on the analytic vortex at sixteen rungs; the classical comparator that actually beats it on the benchmark is named and measured; and the advantage is placed where the same solver meets an operator no transform diagonalises, with the crossover grid, the qubit count and both classical comparators stated as numbers. **Where it is decisive:** perturbation transport and linearised convection on non-uniform mean flows at high Reynolds number and grids beyond 69,000 per axis, where the best classical route needs $10^7\text{--}10^{12}$ s on a core and terabytes of field, and the register stays under 110 logical qubits.

Annex, measured. The tensor-train finite-volume solver on its own merits: memory ratio to the dense field $56\times$ to $92,183\times$ across $\text{Re } 10 \rightarrow 10^4$; wall-clock slower than dense at $\text{Re } \leq 1,000$ and $5.2\times$ faster at $\text{Re } 10^4$ on the same grid at a $2,000\times$ accuracy penalty (airbus_bench_workstation.json); against the spectral floor of the one-mode vortex it stores $18\text{--}105\times$ more (floor_airbus_spectral.json). It is the classical-hardware route to the same operator, not the advantage claim.

Scalability to industrial relevance. The register grows by two grid qubits and two coefficient qubits per doubling of resolution; the operator is a five-point stencil at any N ; the base flow enters through the coefficient oracle with no change to the circuit family; three dimensions add $\log_2 N$ qubits per register and raise the classical exponents by one, which is where the exponent gap widens. The nonlinear extension is Carleman linearisation into the same Schrödingerisation, inside the polynomial bound of the brief's [9].

Business value, bounded. In Airbus's units the value is design iterations per wind-tunnel campaign. As a scenario computed from the table, not a consequence of the model: a linearised-transport case at $262,144^2$ on a non-uniform mean flow at $\text{Re } 10^4$ returns its three design observables in 3.5 years on a 1 MHz fault-tolerant machine against 23 years on one classical core or ≈ 8 days on a 1,000-core node; the fault-tolerant route is the one whose cost rises by $4\times$ per halving of grid spacing while the best classical one rises by $8\text{--}16\times$, and that ratio, not today's constant, is what decides the fleet-scale case.

5. Validation plan

Phase 2, three months, metrics and challengers declared in advance: (1) compile the Hamiltonian-simulation circuit (Trotter and qubitization variants) with the coefficient oracle for $N = 16$ and report T-count and depth, replacing modelled constants with compiled ones and recomputing both crossovers; (2) extend the validation ladder to $N = 256$ and $\text{Re } 10^4$, error and n_p per rung; (3) re-measure both classical comparators on the transport operator at $N = 4,096$ on the workstation, add a semi-implicit multigrid stepper as a third, and the FFT comparator on the benchmark to $N = 16,384$, so every classical cost law carries two more octaves of measurement; (4) pseudo-spectral comparator on the benchmark at matched error, time and memory per rung; (5) re-fly the 42-qubit primitive with the compiled stencil circuits at $N = 4$ on IBM Heron as the hardware execution of one Schrödingerised step. **Acceptance:** compiled constants move the $\text{Re } 10^4$ transport crossover by less than one octave in N with ≤ 120 logical qubits; validation error at every rung equals the finite-volume discretisation error within a factor of two; either outcome is reported.

6. Hybrid / cross-domain integration

The quantum step is one stage of a classical CFD pipeline: mesh, operator assembly and the base-flow field classical; initial-state preparation from the analytic profile; Hamiltonian simulation on the fault-tolerant device; amplitude estimation returns the design observables to the classical post-processor, which never reconstructs the full field. The tensor-train annex is the same operator held classically, so the two routes share stencils, verification and referee; the FFT and Krylov solves are the referees for both.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE): Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (systems), Roshan Ghazali (data), Dr. Ana Sara (validation), Gavin Vaz (platform). Prior work: 42-qubit real-time evolution on IBM Heron with gates declared before the run; the quantum tensor-train finite-volume solver and its Re ladder; 128- and 156-qubit real-time dynamics on two devices in the sister tracks; GIC 2026 dual-track finalist, with the same discipline of measuring the classical wall before placing a claim.

8. Scope, with treatment

(i) On the canonical vortex there is no time-to-solution advantage against the exact FFT solve, which we measured; the benchmark is delivered for validation and scaling, and the advantage is placed on the non-circulant transport operator. (ii) The transport crossover is a resource-model crossing with declared constants against the better of two measured classical cost laws; it is Reynolds-dependent ($65,536$ per axis at $\text{Re } 10^4$, $\approx 10^6$ at $\text{Re } 100$) and against a 1,000-core node it lies beyond 10^6 per axis; Phase 2 replaces the constants with compiled ones and adds a third classical route. (iii) The memory advantage on the one-mode vortex is against grid storage; the spectral representation holds that flow in 128 bytes, and the transport operator's field has no such representation. (iv) The 42-qubit job executes the evolution primitive, not a flow solve. (v) At low Re the diffusive stiffness raises the p register (n_p up to 15 at $N = 128$, $\text{Re } 10$), reported per rung. (vi) Nonlinear flows enter the polynomial bound of the brief's [9]; nothing here claims past it. (vii) The tensor-train annex is faster than dense only at matched grid at $\text{Re } 10^4$, not at matched error.

Appendix A — Hardware job register and receipts

item	device	job ID	receipt
Real-time evolution of a Taylor–Green-prepared state, 42 qubits, 9 steps $\times$ 2 modes, 8,192 shots; waveform correlation 0.993 / 0.971	ibm_marrakesh	d9solsgpdb6s73e6a6q0	qlga_result_20260810_122411.json, qlga_counts_20260810_122411.json, design qlga_design.json
Scout job	ibm_marrakesh	d9solbntfhrrs73dt1t40	qlga_state.json
Schrödingerised solver validation ladder and resource model	CPU	—	schrod_tgv.py, schrod_tgv_results.json, schrod_tgv_fig.png
Exact FFT solve of the benchmark operator to $N = 4,096$ (the classical wall on the canonical vortex)	CPU	—	floor_airbus_fft_exact.py, floor_airbus_fft_exact.json, FLOOR_AIRBUS_FFT_EXACT.md
Krylov solve of the transport operator to $N = 512$ and the crossover model	CPU	—	airbus_noncirculant_crossing.py, airbus_noncirculant_crossing.json, AIRBUS_CROSSING_NONCIRCULANT.md
Pseudo-spectral time-stepper on the transport operator to $N = 1,024$ (agrees with Krylov to 2×10^{-6}); best-of-two summary	CPU	—	airbus_pseudospectral_comparator.py, airbus_pseudospectral_comparator.json, airbus_crossing_summary.json
Tensor-train Re ladder (N , steps, χ , bytes, L2, wall)	CPU	—	qtt_ladder.json; Re 100 re-run results/rerun_Re100_20260914.json

item	device	job ID	receipt
Dense vs tensor-train wall-clock and error, Re $10 \rightarrow 10^4$	CPU, workstation	—	airbus_bench_workstation.json
Spectral storage floor on the canonical flow	CPU	—	floor_airbus_spectral.json

Appendix B — Why the linear operator is exact for the benchmark, and why it is circulant

For $u = U_c - \cos(x - U_c t) \sin y \cdot e^{\{-2vt\}}$, $v = \sin(x - U_c t) \cos y \cdot e^{\{-2vt\}}$, the convective term $(u \cdot \nabla)u$ equals $-\nabla(\frac{1}{4}e^{\{-4vt\}}(\cos 2(x - U_c t) + \cos 2y)) + U_c \partial u$: a gradient absorbed by the pressure plus mean-flow advection. What remains is $u_t + U_c u_x = v \nabla^2 u$ per component. On the periodic cell grid with central fluxes its matrix $A = -U_c D_x + vL$ is a sum of shift operators with constant coefficients, hence circulant in both axes and diagonal in the two-dimensional discrete Fourier basis: $\exp(AT)u$ is one forward transform, one diagonal multiply and one inverse transform. The Schrödingerised solver in the package simulates its own output in exactly that basis, which is how the package's validation column is computed; the floor receipt reproduces that column to four digits. A non-uniform base flow multiplies the shift operators by a position-dependent field, breaks the circulant structure, and removes the transform route; that is where §4's crossing sits.

Appendix C — Claim ledger (measured · modelled · comparator · quantum attribution · cost · acceptance)

claim	status	classical comparator	quantum attribution	cost charged	acceptance
Solver reproduces the analytic vortex to discretisation error, sixteen rungs	measured (exact simulation, 16–29 qubits)	exact classical solve of the same operator; analytic solution	algorithm output, simulated	minutes CPU	error within $2 \times$ of FV at every rung
No time-to-solution advantage on the canonical vortex	measured	exact FFT solve, 6.5 s at $N = 4,096$	—	seconds CPU	stated, not contested
Memory $2 \cdot \log N + n_p$ qubits vs grid storage	exact	grid storage; spectral floor named	register size	—	—
Transport-operator crossover at $N \approx 65,536$ (Re 10^4) and $\approx 10^6$ (Re 100), single core	classical side measured to $N = 512$ (Krylov) and 1,024 (stepper) and fitted; quantum side modelled, constants declared	best of sparse Krylov and pseudo-spectral stepper	Hamiltonian simulation with coefficient oracle	readout $O(1/\varepsilon)$ and oracle $\times 2$ charged	compiled constants move the Re 10^4 crossover < 1 octave at ≤ 120 logical qubits (Phase 2)
42-qubit evolution primitive on hardware	measured	exact statevector of the medium	hardware execution	1 banked job	correlation ≥ 0.9 after one envelope (met)
Tensor-train annex: Re ladder, memory ratio vs dense	measured	dense FV; spectral floor	quantum-inspire d	workstation hours	$L2 \leq 10^{-4}$ per rung (met)